When can a prediction replace a physical trial? A production-referenced evaluation of design-stage decision rules in deep drawing

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Abstract

Simulation and machine learning inform design-stage manufacturability decisions, yet they are judged by prediction error, not by whether their verdicts could replace a physical trial. This paper proposes a production-referenced evaluation of design-stage decision rules. Its unit, the within-run floor, is the difference that drift and scatter of production and measurement produce between two batches of one design within a run. For each evidence relation of a new design to produced ones (a sibling, a new process setting or a new family), a rule that returns meets, fails or trial is characterised by its decisive distance, beyond which at least 95 % of its verdicts are correct, and its safe distance, beyond which at most 5 % are wrong; a guard band turns them into single verdicts. An exploratory application to one laboratory deep-drawing process, with alternatives produced once and an uncalibrated simulation dataset, scored simulation-based, learned and production-record rules. The nearest produced setting, committed to in advance, decided at 3.4 floors with a sibling and 8.5 for a new process setting. No simulation-based or learned rule was more decisive; a Gaussian process specified for small data reduced to a force trend. Intervals calibrated on the same relation in the other family, not learning, made the production record safe for new settings. For a new family only simulation-based rules decided, none closer than 30 floors. Within a family, a production record indexed by its relation to the coming design is the baseline a learned manufacturability check has to beat.

Keywords: decision support, design for manufacturing, machine learning, uncertainty quantification, selective prediction, deep drawing

1 Introduction

Whether a design alternative can be made to its requirements is decided long before it is made. In sheet-metal forming the decision rests on a finite-element simulation (Banabic, 2010): the team compares the predicted characteristics with the requirements and releases the alternative, changes it or sends it to a physical trial. Artificial intelligence makes this loop faster: surrogates replace simulations, models learn manufacturability from earlier and synthetic designs (Yan et al, 2026; Ghadai et al, 2018; Ballegeer and Benoit, 2026), and predictive-quality models and virtual metrology estimate quality from process data (Tercan and Meisen, 2022; Dreyfus et al, 2022). Such models are judged by their prediction error on held-out data, which leaves open the question the team has to answer: *can this prediction be relied on to decide, or is a trial still needed?* Mechanical design cannot be verified by simulation the way integrated circuits are (Whitney, 1996), and the answer depends on how much production has been seen and how the new design relates to it.

Uncertainty quantification supplies intervals (Nemani et al, 2023), and virtual metrology attaches a reliance index to each prediction (Cheng et al, 2008), but neither says how close to a requirement a verdict can be relied on, in a unit that production defines, for each relation of a new design to the produced evidence. This paper asks three questions. How can the fitness of a design-stage prediction to decide be expressed on a production-referenced scale, for any predictor and characteristic (RQ1)? How does it depend on the amount of production evidence and on how the new design relates to it (RQ2)? And, with six to nine produced alternatives, what do a generic simulation dataset, Gaussian processes, gradient boosting and calibrated intervals contribute against a production-record baseline (RQ3)?

The proposed framework has the *within-run floor* as its unit: the difference that drift and scatter of production and measurement produce between two batches of one design within a run. A rule that turns the design-stage evidence into a verdict (*meets*, *fails* or a physical *trial*) is scored, for each *evidence relation* between a new design and the produced alternatives, by two points on its operating characteristic, its *decisive* and *safe distances*; a procedure turns them into a guard band for single verdicts and prices them by the costs of errors and trials. The decisions in scope are those of process planning, the approval of process changes in series production and the qualification of AI-based tools that support them, taken by a team that has produced variants of a design family; the tryout of a new die, in which the tool itself changes between loops, is not.

The application uses open data: 9,000 deep-drawn and cut cups of dual-phase steel, produced as 18 design–process alternatives of 500 consecutive parts each on one laboratory tool set (Baum and Heinzelmann, 2026), with finite-element simulations of the same process from a generic dataset (Baum and Heinzelmann, 2025). Each alternative was produced once, and the simulations were not calibrated to the process. The study is exploratory: it shows whether the framework discriminates between rules and evidence situations on real data, and its findings are hypotheses for confirmation. The contributions are the framework (Section 3); an evaluation design that separates evidence relations, isolates what the simulation adds and tests the learners against

a production-record baseline (Section 4); findings for this process (Section 5); and a proposed reporting checklist with an open scoring function (Section 6.4).

2 Background and related work

2.1 Machine learning for manufacturability, predictive quality and virtual metrology

Design for manufacturing codifies what a process can make into rules and cost models (Boothroyd et al, 2010); machine learning learns manufacturability from CAD models (Ghadai et al, 2018; Zhang et al, 2018), synthetic sheet-metal datasets (Ballegeer and Benoit, 2026), process data and simulated design spaces (Yan et al, 2026). In production, predictive-quality models and virtual metrology estimate quality from process data (Wuest et al, 2016; Tercan and Meisen, 2022; Dreyfus et al, 2022); their uncertainty is quantified by Bayesian, ensemble and conformal methods (Nemani et al, 2023; Heddoub et al, 2026). Automatic virtual metrology goes further and attaches to every prediction a reliance index and a global similarity index that flags process data unlike the training data (Cheng et al, 2008, 2012), the closest analogue in this literature to the guard band and the applicability guards of this paper. Models are carried across products and processes by transfer learning and domain adaptation (Maschler and Weyrich, 2021), also from simulation to real parts (Tercan et al, 2018). In sheet forming, machine learning predicts defects and springback from simulated and measured data (Cao et al, 2024; Heinzelmann et al, 2025b), and discrepancies between synthetic and real data are being addressed for deep drawing (Baum et al, 2026b). These models are evaluated by held-out accuracy or coverage, and language models by case studies (Makatura et al, 2024) and benchmarks (Picard et al, 2025), not by their decisions against the production of the alternatives they judge.

2.2 Simulation credibility and the validation of forming simulation

Verification and validation ask whether a model is accurate enough for its intended use, distinguishing its validation from its application domain (Oberkampf and Roy, 2010); credibility assessment scales the required evidence with the consequence of a wrong decision (ASME, 2018). Bayesian calibration estimates calibration parameters together with a model discrepancy (Kennedy and O'Hagan, 2001; Higdon et al, 2004), without which calibrated parameters are biased and over-confident (Brynjarsdóttir and O'Hagan, 2014; Arendt et al, 2012); multi-fidelity models combine a cheap and an expensive model through a fitted scale and a discrepancy (Kennedy and O'Hagan, 2000; Le Gratiet and Garnier, 2014). In sheet forming, the friction law (Hol et al, 2012), the unloading and hardening behaviour that govern springback (Yoshida and Uemori, 2002; Wagoner et al, 2013) and the elastic deflection of the tools (Pilthammar et al, 2018) are the main sources of simulation error; robustness analysis propagates the scatter of material and process inputs (Wiebenga et al, 2014; Chen et al, 2022), and draw-in is measured and controlled in line in series production (Havinga et al, 2020; Fischer et al, 2018). Deep-drawing simulation datasets support surrogate and

transfer learning (Baum and Heinzelmann, 2025; Heinzelmann et al, 2025a), and the simulation-to-part deviation has been decomposed statistically (Baum et al, 2026a); how close to a requirement a calibrated or learned prediction can decide remains open.

2.3 Decisions with abstention and their evaluation

In conformity assessment a measured value proves conformance, proves non-conformance or leaves the decision open, with guard bands set by the measurement uncertainty (ISO, 2017); the global consumer's and producer's risks follow from a prior distribution of the measurand (Joint Committee for Guides in Metrology, 2012). In acceptance sampling and selection procedures the operating characteristic gives the probability of a decision against the true quality, with an indifference zone inside which no decision is required (Bechhofer, 1954). A classifier with a reject option trades errors against abstentions (Chow, 1970; Geifman and El-Yaniv, 2017), as does regression with a reject option (Zaoui et al, 2020); learning to defer passes difficult cases to an expert (Madras et al, 2018). Cost-sensitive evaluation (Elkan, 2001) and decision curves (Vickers and Elkin, 2006) price the errors, and appropriate reliance on automation requires that its reliability be visible to its users (Lee and See, 2004), though displayed confidence can calibrate trust without improving joint accuracy (Zhang et al, 2020, 2021). The decisive and safe distances are operating characteristics of this kind, and the guard band is the design-stage counterpart of a conformity-assessment one, whose dependence on a prior of requirement distances mirrors that of the global risks.

2.4 Production variation, measurement and design decisions

Statistical quality control compares short- and long-term variation with tolerances through capability and performance indices, with recommended minima for existing processes (Montgomery, 2019); in an initial process study an index above 1.67 meets the acceptance criteria and one between 1.33 and 1.67 may be acceptable after a review by the customer (Automotive Industry Action Group, 2006), and time-dependent models describe drifting processes (ISO, 2026). Precision experiments define the repeatability limit, the difference that two results under repeatability conditions exceed with a probability of 5 % (ISO, 2025), and intermediate precision under conditions that differ in time (ISO, 2023); measurement systems analysis separates repeatability from bias, linearity and stability (Automotive Industry Action Group, 2010). In engineering design, decisions between alternatives with uncertain outcomes (Hazelrigg, 1998) are taken with margins (Eckert et al, 2019) or with tests whose cost is traded against that of errors (Thomke and Bell, 2001; Salado and Kannan, 2018), and what a team knows about a new alternative depends on whether it is a variant, an adaptation or an original design (Pahl et al, 2007), of which the evidence relations are the counterparts for production evidence.

Table 1 places the framework among these methods. It adopts the indifference zone of operating characteristics, the guard band of conformity assessment, the selective risk of abstaining predictors and grouped cross-validation (Roberts et al, 2017). What is new is the production-defined unit in which every rule and characteristic is scored,

Table 1 Positioning among methods that support or evaluate design-stage manufacturability decisions

Approach	Evidence	Output	Scored against production	Abstains
DFM rules and cost models (Boothroyd et al, 2010)	codified process knowledge	guideline compliance, cost	no	no
V&V, credibility assessment (Oberkampf and Roy, 2010; ASME, 2018)	physics model, validation tests	accuracy; required evidence	on test cases	can require tests
Bayesian calibration (Kennedy and O'Hagan, 2001)	simulations, experiments	predictive distribution	not as decisions	implicitly
Conformal prediction (Lei et al, 2018)	exchangeable calibration data	interval with coverage	coverage	yes
Virtual metrology with reliance index (Cheng et al, 2008)	process data of produced items	value, reliance and similarity index	on held-out production	flags
Conformity assessment, OC curves (ISO, 2017; Bechhofer, 1954)	measurement and its uncertainty	accept, reject, undecided; risks	on the measured item	yes
Selective classification and regression (Chow, 1970; Zaoui et al, 2020)	labelled data	value or reject; risk–coverage	on held-out data	yes
ML manufacturability models (Ghadai et al, 2018; Yan et al, 2026)	historical or simulated designs	class or value	on held-out data	rarely
This paper	simulations, produced alternatives	verdict; distances in floors by evidence relation	yes, per verdict	yes

the qualification by evidence relation, and their combination into a procedure for design-stage decisions that any predictor can enter.

3 The framework

3.1 The decision problem, its scope and its loss

A design alternative a is a combination of product geometry and process settings produced as a series of parts, over which a quality characteristic c varies. A requirement is an upper limit R on the characteristic, or on its absolute value for characteristics whose target is zero, that a share p of the parts must meet: alternative a is *adequate* for c when the p-quantile of its parts satisfies $q_p(a, c) \leq R$. At the design stage q_p is unknown. A decision rule r maps the available evidence to an interval $[L_r, U_r]$ for q_p and to a verdict

$$V_r = \begin{cases} \text{meets} & U_r \leq R, \\ \text{fails} & L_r > R, \\ \text{trial} & \text{otherwise.} \end{cases} \tag{1}$$

The three verdicts mirror the decision rules of conformity assessment (ISO, 2017). A rule that returns a point ($L_r = U_r$) always decides. The evidence consists of simulations of a and of the production record of other alternatives, the *calibration alternatives*. The verdict costs c_{FA} when a failing alternative is accepted, c_{FR} when an adequate one is rejected and c_{T} when it is sent to trial, so a rule's expected cost per verdict is

$$C = c_{\mathrm{FA}} P_{\mathrm{FA}} + c_{\mathrm{FR}} P_{\mathrm{FR}} + c_{\mathrm{T}} P_{\mathrm{T}}, \tag{2}$$

which the two distances of Section 3.4 summarise at a fixed share of errors. Three limits of scope apply: the demonstration uses upper limits on the 95th percentile ($p = 0.95$, for a normal characteristic a performance index of about 0.55), lower limits following by a change of sign; verdicts are per characteristic, a design-level verdict over all characteristics being reported as an extension (Section 5.7); and attribute outcomes such as splits and wrinkles, which need a requirement on a rate, are not covered.

3.2 The within-run floor

Production evidence comes as series of consecutive parts, each cut into batches of B consecutive parts summarised by a robust centre m (here the 10 % trimmed mean). The within-run floor of characteristic c, the floor for short, is

$$F_c = Q_\alpha\big\{\, |m_i - m_j| : i < j \text{ batches of the same alternative} \,\big\}, \tag{3}$$

the α-quantile of the difference between two batches of one design. Under a stationary normal model with a standard deviation σ_{b} between batch means and σ_{w} between parts, $F_c = 1.96\sqrt{2}\sqrt{\sigma_{\mathrm{b}}^2 + \sigma_{\mathrm{w}}^2/B}$ at $\alpha = 0.95$, the form of the repeatability limit (ISO, 2025, 1994); because it pools batch pairs up to 450 parts apart and grows with their lag, it is closer to a time-different intermediate-precision limit (ISO, 2023) of batch centres within a run. The reference protocol is $B = 50$, ten batches of a 500-part run, all pairs of batches and $\alpha = 0.95$; every characteristic is scored in the floor of its own geometry, and the pooled floor serves only to convert distances to physical units. It rests on three assumptions: batches of one design differ only by drift and scatter within a run (A1); future runs vary within themselves as the runs it is estimated from do (A2); and the measuring system is repeatable, its resolution small against the floor (A3).

Four properties matter for its use. It has the unit of the characteristic, so distances in floors put all characteristics and predictors on a common scale, though equal distances do not mean equal nonconforming fractions. It is a property of a protocol, and σ_{b} and σ_{w} are archived so that it can be recomputed. It contains whatever drifts between batches, process and measuring system alike: without a reference artefact scanned through the series it is a production-and-measurement floor, and not the shop floor. And it excludes the variation between runs, which makes it a lower bound of the variation of series production. A new family, whose floor is unknown before it is produced, is scored in the floor of the produced family. Two alternatives a and b are resolvable for c when their effect-to-scatter ratio $\mathrm{ESR}_c(a, b) = |\bar{m}_a - \bar{m}_b|/F_c$ is at least

one, $\bar{m}$ being the centre of all parts of an alternative; for a continuous parameter x with local sensitivity $s_c = \partial\bar{m}/\partial x$ the minimum resolvable change is $F_c/|s_c|$.

3.3 Decision rules

Eight rules carry the main results (Table 2). The nominal simulation (M0) is the simulation at nominal sheet thickness and friction. The scaled simulation (M1s) fits $q_p = a + b\,s_0$ to the calibration alternatives by least squares, the scale term of a multi-fidelity correction without its discrepancy. The envelope rule (M2) adds the median offset of the calibration alternatives and their largest absolute residual to the top of the simulation envelope over the dataset's thicknesses and friction coefficients. The GP-corrected envelope (M5) models the offset between q_p and the envelope as a Gaussian process (Rasmussen and Williams, 2006) fitted by maximum marginal likelihood, its noise variance bounded below by the variance between the calibration series of one geometry and force (the noise-variance bound). The nearest produced setting (NN) is the mean q_p of the calibration alternatives at the nearest force, a capability-database look-up, and the force trend (M1sn) the least-squares line of q_p in force, the counterpart of M1s without the simulation, which within a geometry is a function of the force.

The trend GP (M7n) is a learner specified for six to nine produced alternatives: a Gaussian process of q_p with a linear mean in force and lubrication whose coefficients are integrated out (universal kriging), weakly informative priors on length scales and noise instead of hard bounds, and a noise term made of the known sampling variance of each q_p and an estimated between-run nugget that absorbs no modelled effect. Its counterpart with the simulation (M7) is the autoregressive multi-fidelity model (Kennedy and O'Hagan, 2000), with the simulation in the mean and a GP discrepancy. Finally, intervals are separated from predictors: P+RM puts around the prediction of a rule P a relation-matched interval, the 90 % split-conformal quantile (Lei et al, 2018) of P's absolute errors in floors over the cases of the same relation in the other family, a group-conditional construction (Vovk et al, 2005). The errors are borrowed because a family's own calibration set has no case of its new-setting relation, so the intervals guarantee coverage only if the two families' errors are exchangeable, which the data cannot establish (Barber et al, 2023); NN+RM is NN with such an interval.

Ten further rules complete the comparison. M1 adds the median offset of the calibration alternatives to M0, and M1n is their median q_p; M2n and M5n are M2 and M5 without the simulation. Mc fits one friction coefficient per lubrication pattern to the draw-ins by least squares, without a discrepancy, and Mcd calibrates the friction with a GP discrepancy (Kennedy and O'Hagan, 2001) and predicts by the posterior mixture over the friction grid. M3 corrects the matched simulation of every part by gradient boosting, and M3n models the characteristic by boosting without the simulation, both with jackknife+ intervals; M0w is the range of the simulation over thickness and friction; and M6, a robustness variant, puts a conformal margin on standardised residuals around the centre of M5.

Table 2 The rules of the main results; the further rules are defined in Section 3.3

Rule	Prediction of q_p	Interval (level)	Evidence
M0 nominal simulation	simulation at nominal thickness and friction	point	none
M1s scaled simulation†	$a + b$ M0, least squares	point	q_p
M2 envelope + residual margin	envelope top plus the median offset	largest residual $(\frac{n-1}{n+1})$	q_p
M5 GP-corrected envelope	envelope plus a GP offset, ML-II	GP (nominal 0.9)	q_p
NN nearest produced setting†	mean q_p at the nearest produced force	point	q_p
M1sn force trend†	$a + b$ force, least squares	point	q_p
M7n trend GP‡	linear mean plus GP, MAP under priors	GP (nominal 0.9)	q_p
NN+RM‡	NN	relation-matched (0.9)	q_p

Fitted on the calibration alternatives; descriptors: geometry indicator, blank-holder force and lubrication rank. Pre-specified unless marked: † added after the first results; ‡ added for this version. Levels: guaranteed (largest residual, n calibration alternatives: 8 with a sibling, 6 for a new setting, 9 for a new family) or nominal.

3.4 Operating characteristics: decisive and safe distances

A rule is scored by judging each produced alternative as a new design at requirements placed at a known distance from the truth,

$$R = q_p(a, c) + d\,F_c, \tag{4}$$

so that a is adequate exactly when $d \geq 0$. Let $\kappa(\delta)$ and $\omega(\delta)$ be the shares of correct and of wrong verdicts among all feasible verdicts at $|d| = \delta$: every case gives one verdict on an adequate alternative and, where the requirement is not negative, one on a failing alternative. The *decisive distance* δ_d is the smallest δ beyond which $\kappa \geq 0.95$, and the *safe distance* δ_s the smallest δ beyond which $\omega \leq 0.05$ (Fig. 1a). In plain terms, for a rule that always decides, $\delta_\mathrm{d} = \delta_\mathrm{s}$ is close to the 90th percentile of its absolute error in floors; an interval rule trades decisiveness for safety, and between δ_s and δ_d it sends designs to trial rather than misleading. Both are points on operating characteristics over a population of design situations; δ_d is the analogue of the indifference zone of a selection procedure (Bechhofer, 1954) at a probability of correct selection of 0.95.

The distances are exact up to the 0.05-floor grid on which they are evaluated, because κ and ω change only where δ crosses the distance by which a case's interval excludes its truth. They are order statistics, so a few verdicts far out can move them: with 126 cases, some 240 verdicts, the 95 % threshold rests on about a dozen. Distances below about one floor are reported as "<1", since production reproduces the decided quantile only to about one floor within a run (Section 5.1). The distances weigh false accepts and false rejects equally, and as δ grows, failing-side requirements become infeasible for characteristics close to zero, so the failing side carries half the weight up to a few floors and a third at about 100 floors; the safe distances on the false-accept and the false-reject side are therefore reported as well.

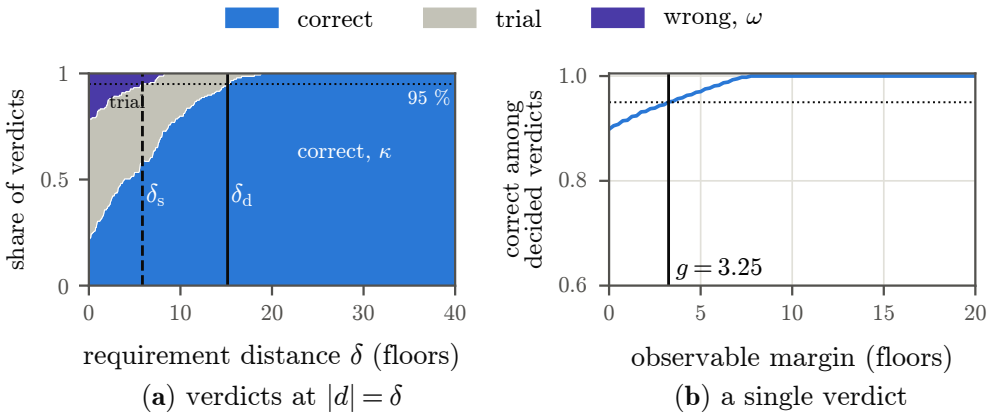


Fig. 1 Operating characteristics of one rule (M7n, extrapolated setting): verdict shares against the requirement distance, with δ_d and δ_s (a), and correct share among decided verdicts against the margin, with g (b)

3.5 Evidence relation and calibration budget

How a new design relates to the produced alternatives is part of the decision situation. With a *sibling*, an alternative at the same process setting has been produced that differs only in the factor varied at a fixed setting (here the lubrication pattern, which the nominal simulation does not represent); a *new process setting* lies between produced settings (interpolation) or outside them (extrapolation); a *new family* shares no produced alternative. A team knows, before it decides, whether the produced siblings of a new variant differ resolvably from each other, so the sibling relation is stratified by this decision-time information, not by the new design's own production. The calibration budget is the dependence of the distances on the number k of produced alternatives. An applicability guard flags a design outside the calibration, in the spirit of a valid input domain (Malak and Paredis, 2010): its geometry is not calibrated, a descriptor lies outside the calibrated range, or its simulated q_p lies outside that of the calibration alternatives.

3.6 Procedure

Figure 2 turns the framework into six steps, steps 1 to 5 once per design family and repeated when production evidence is added, and step 6 for every new design. Steps 1 to 4 frame the decision (family, alternatives, characteristics, form of the requirement, costs), measure the floor with a stated protocol, state which differences production resolves, and calibrate the candidate rules with and without the simulation.

Step 5 qualifies the rules of the family by their distances for each evidence relation and fixes for each rule and relation a *guard band* g: the smallest observable margin between the predicted interval and the requirement beyond which at least 95 % of the decided verdicts are correct. Being conditional on what is observed, g depends on the distribution of requirement distances that the team expects, its prior, as the global risks of conformity assessment do (Joint Committee for Guides in Metrology, 2012).

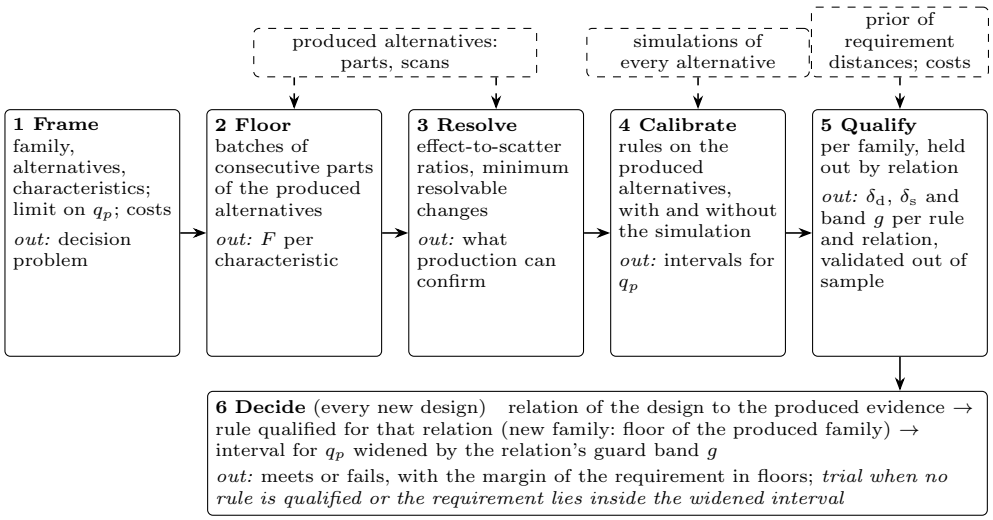


Fig. 2 The procedure: steps 1–5 once per design family, step 6 for every new design; dashed boxes are the evidence and inputs it uses

The reference prior weights the 47 distances of the evaluation grid within ±20 floors equally (0, ±0.5 to ±10 in steps of 0.5, ±12, ±15, ±20), requirements set a few to some tens of floors from the true 95th percentile (±20 floors are about ±1 mm of mid-side draw-in, ±1.6° of wall angle, ±0.2 mm of cup depth and ±0.07–0.16 mm of dome); a release-like prior, the grid within ±5 floors, stands for requirements near capability limits, and a team replaces both by the distribution of its own requirements, for example the margins between past drawing limits and achieved 95th percentiles. The band is estimated per family and validated out of sample, leaving out one alternative and, for new settings, one force cell.

Step 6 establishes the relation of the new design, applies the rule qualified for it, widens the rule's interval by g floors and returns meets or fails with the margin in floors, or a trial when no rule is qualified or the requirement lies inside the widened interval. In pseudocode, step 5 is: for each rule and relation, collect the held-out verdicts over the prior; for $g = 0, 0.25, \ldots, 20$, compute the share correct among verdicts decided with a margin of at least g; take the smallest g beyond which it stays at or above 0.95. Step 6 is: $V =$ meets if $U + gF \le R$, fails if $L - gF > R$, trial otherwise. The band of the relation takes the place of the range and family guards, which here coincide with the extrapolated and new-family relations.

3.7 A worked example

A process planner has to decide whether the concave cup can be run at 300 kN with medium lubrication, a setting not yet produced, without a trial run, the six alternatives at 100 and 500 kN having been produced. Step 1 frames the decision: the characteristic is the mid-side draw-in, by which the blank-holder force is set, the requirement (chosen for illustration) that 95 % of the parts draw in by at most 15.30 mm, and a trial

costs a short press run, about a twentieth of a false accept. Step 2 gives the floor from the produced series, 0.050 mm, and step 3 shows that force changes of 21–67 kN are resolvable. Step 4 calibrates the rules on the six alternatives. Step 5 qualifies the nearest produced setting for interpolation: it decides at 4.7 floors and needs a guard band of 1.5 floors on this family (2.0 pooled over the families); out of sample, 95 % of its decided verdicts stayed correct. Step 6: the nearest setting, the mean of the six alternatives at the neighbouring forces, predicts 14.99 mm, a margin of 6.2 floors, more than its band, so the planner releases the setting. The trend GP agrees with a margin of 4.5 floors; M2, M5 and M5n, widened by their bands, contain the limit and would send the design to trial, and the nominal simulation, qualified for no relation, predicts 15.66 mm. The 500 parts had a 95th percentile of 14.88 mm, 8.4 floors below the limit.

4 Demonstration on deep drawing

4.1 Alternatives and production data

The production data are the Real Deep Drawing and Cutting dataset (RDDAC) of the University of Stuttgart (Baum and Heinzelmann, 2026; Baum et al, 2026a). Square blanks of DP600 dual-phase steel, 210 mm wide and about 1 mm thick, were deep drawn into cups (OP10) and cut (OP20) on an industrial press with one tool set, whose two die geometries, concave and convex side walls, differ in base curvature, wall angle and bottom radius at once. Each geometry was run at three blank-holder forces (100, 300 and 500 kN) and three lubrication patterns applied to the blank (coarse, medium and fine, with series medians of 1.15–1.48, 0.92–1.23 and 0.85–1.13 g/m^2 of oil): 18 alternatives, each produced as one series of 500 consecutive parts. Every part has the signals of four load cells around the die at 300 Hz, the punch position and temperature, sheet-thickness and oil-film traverses and laser scans after both operations. The series warm up by 1.0–5.0 K over their first 150 parts from punch temperatures of 20.5–29.6 °C, their median sheet thickness is 0.986–0.993 mm, and the 100 kN series ran at 47 mm/s against 78–79 mm/s, so the extrapolation to 100 kN also changes the sliding speed and stroke rate on which the lubricant film depends. The experiments were collected between September and November 2025, in contiguous blocks of 500 per alternative; the documentation states neither the order of the blocks nor whether the parts were scanned in production order, and gives no material certificate, coil record, tool drawing, press or cushion type, or method of applying the lubrication patterns.

4.2 Simulations

The simulations come from the Deep Drawing and Cutting Simulations dataset (DDACS) (Baum and Heinzelmann, 2025; Heinzelmann et al, 2025a), a generic dataset of some 32,000 LS-DYNA simulations of the same cup family that was neither set up for nor validated against the produced process. Its documentation gives rigid tools (blank holder, die, punch), shell elements with seven integration points through the thickness and nodes about 1 mm apart, a scaled DP600 flow curve, constant Coulomb friction, a drawing depth of 30 mm, trimming in OP20 without modelled cutting tools, and springback after both operations; it leaves open the element formulation, whether

contact accounts for the shell thickness, the yield criterion and hardening law, the blank-holder actuation and the springback procedure, and it does not represent the stroke speed. For the RDDAC tools, 396 simulations vary the sheet thickness (0.95–1.00 mm) and the friction coefficient (0.05–0.15) at each geometry and force, all with the nominal flow curve. A part is matched to the simulation with the nearest thickness and the friction coefficient that the dataset's linear mapping assigns to its oil film, an assumption of the dataset's authors on which M3, M3n and the matched simulations depend. The nominal simulation is at 0.99 mm, nearest the median thickness, and friction 0.10, the middle of the grid; Mc and Mcd calibrate the friction.

4.3 Measurement and quality characteristics

The parts were measured with a single-sided laser line scanner, which records a height and a luminescence image of 3200 by 2000 points, 0.077 mm apart along the laser line (x) and 0.158 mm apart between lines in the direction of motion (y), with height steps of 7.75 µm, the cup bottom facing the scanner (Fig. 3). The scales are not traced to the part: the lateral one comes from the scanner specification and a 30 mm block, the line spacing from the scans, partly fitted to the simulation, and the height from a 1 mm block, against a nominal repeatability of 5 µm. Absolute values are therefore uncalibrated, which limits every absolute comparison with the simulation but not the floor, a within-alternative difference that a constant scale error only rescales.

Seven characteristics were extracted with the same definitions from the scans and from the simulated nodes, referred to the plane of the cup bottom (Table 3): the mid-side and corner draw-in and the flange waviness, process indicators on the flange that the cut removes; the wall angle and the cup depth, measured after drawing because the cut removes the flange that defines the depth and the cut walls facing the scanner read several degrees too steep; and the arm angle and bottom dome of the cut part, the last two with requirements on absolute values. The four sides of a part measure several of them twice, which exposes direction-dependent systematic errors of up to 3.5 mm in flange width and 10° in wall angle (Table 3); the directions kept avoid the worst of them, but their residual bias is unknown. The mid-side draw-in is quantised in steps of 0.019 mm (0.38 floors), so that a series takes only 6 to 21 distinct values (Fig. 4).

4.4 Evaluation design

Every produced alternative is judged as a new design under grouped cross-validation (Roberts et al, 2017), one scheme per evidence relation. *New variant*: the alternative alone is held out and the other eight of its geometry calibrate, two siblings at its force among them. *New process setting*: the three alternatives of one force of a geometry are held out, the six others calibrate; 300 kN is interpolated, 100 and 500 kN extrapolated. *New family*: one geometry calibrates the rules for the other (two transfers); a few-shot variant adds one to three produced alternatives of the new family. Every calibrated rule is evaluated with and without the simulation (M1s/M1sn, M2/M2n, M3/M3n, M5/M5n, M7/M7n). Each relation has 126 cases (18 alternatives, 7 characteristics), but the units of replication are the 18 alternatives, whose characteristics are correlated.

Table 3 The quality characteristics and their measurement by the laser scanner

Characteristic	Op.	Region; definition	Step/F	Values	Direction-dependent differences
Draw-in, mid-side	OP10	flange mid-sides; $(210 - W_x)/2$, W_x along the laser line	0.38	6–21	$W_y - W_x$ 2.29–3.47 mm; W_y, in the direction of motion, scatters 3.7–17.6 times more
Draw-in, corner	OP10	flange corners; mean of the two diagonals	<0.01	491–500	the two diagonals differ by 1.64–2.17 mm
Flange waviness	OP10	flange; residual SD about a quadratic surface	<0.01	469–494	–
Wall angle	OP10	side walls at mid-depth; mean of east and west	<0.01	497–500	convex: north wall −1.64 to −0.92° from east/west (concave −0.05 to +0.39°); south wall shadowed, 6.7–10.0° off; after the cut, north 3.3–5.6° off (convex)
Arm angle, cut	OP20	cut arms, centre lines; mean of the four arms	<0.01	496–500	–
Cup depth	OP10	flange above the bottom plane; median	<0.01	486–497	the flange plane is cut away at OP20
Bottom dome, cut	OP20	bottom sag at $r = 40$ mm; paraboloid fit	<0.01	474–497	–

Op.: operation measured (OP10 after drawing, OP20 after cutting). Step/F: median step between adjacent values in one series over the floor; Values: distinct values per series. Differences from the redundancy of the four sides, ranges of the series medians; the directions used avoid them, their residual bias is unknown. Absolute values are not calibrated against a reference at the size of the part.

The uncertainty of the distances is described in three ways. The intervals of Table 5 resample the held-out alternatives within each geometry 1000 times (those held out together for a new setting individually), each resample with a block-bootstrap replicate of every true q_p (non-overlapping blocks of 50 parts), the fitted rules held fixed; common to all rules, they give paired intervals of differences, descriptive and not adjusted for the many comparisons. A refitting bootstrap redraws the three lubrication patterns of each geometry with replacement and refits every alternative-level rule, which shows the effect of losing a pattern rather than sampling uncertainty. Coverage gets cluster-bootstrap intervals over alternatives. Each truth is the 95th percentile of a single run, so a sensitivity analysis adds to every truth a between-run component of standard deviation τ floors, from near-replicate reproducibility up to the variation between series. The selection of rules and bands is cross-fitted between the geometries, chosen on one and evaluated on the other; like any cross-validated estimate, the distances describe the fitting procedure on this design rather than one fitted rule (Bates et al, 2024).

4.5 Computation and development record

The constants of the floor and of the requirement grid were set before the decision results were seen, which the archive cannot verify, as no timestamped snapshot was

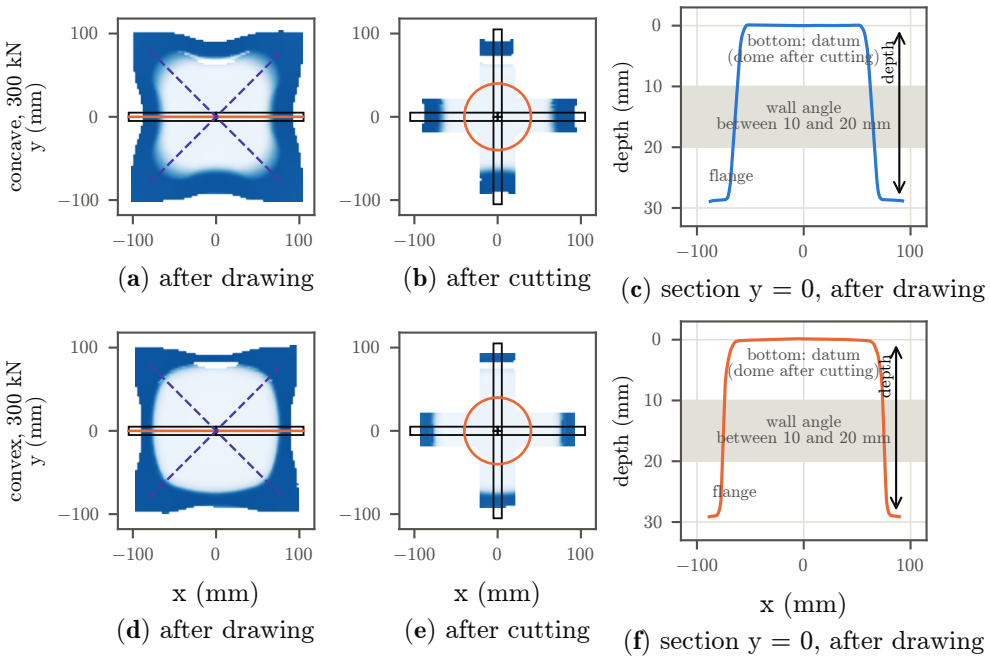


Fig. 3 Scanned parts after drawing and cutting, depth below the cup bottom (a, b, d, e), and sections (c, f); lines and boxes: directions and regions of the characteristics

deposited at the time. The rules were not fixed in advance: M0–M5 and M3n were defined first, and everything else was added after the first results, most of it in response to reviews of earlier versions simulated with a large language model in four rounds (M0w, Mc, M1n, NN, M2n, M5n, the grouped schemes, the exact estimator, the noise-variance bound, M1s, M1sn, the jackknife+ intervals, the guard bands with their priors and the new family scored in the produced family's floor). A fifth simulated round, of the version prepared for this journal, led to M7n, M7, Mcd, the relation-matched intervals, the decision-time strata, the per-family and out-of-sample guard bands, the costs, the between-run sensitivity, the cross-fitting, the few-shot transfer and the design-level verdicts. The comparison of rules is therefore exploratory and carries the optimism of a selection made on the evaluation data (Cawley and Talbot, 2010; Varma and Simon, 2006), which the cross-fitting bounds; every variant tried is reported in the paper or in the archived results (Canseven, 2026), in which `results/findings.md` maps every number of the paper to its script and result file.

Computation. M5 and M5n are Gaussian-process regressions (constant times anisotropic squared-exponential kernel plus white noise, length scales in $[0.1, 100]$, maximum marginal likelihood) on a geometry indicator, the force in 100 kN and the lubrication rank; the noise-variance bound is the larger of the mean sampling variance of the calibration truths and the pooled variance between their series of one geometry and force. M7n and M7 work in pooled floors, with log-normal priors whose medians are 200 kN and one pattern step for the length scales, two floors for the GP and one floor for the nugget standard deviation (log-scale standard deviations 1, 1.5 and 1).

Mcd has a uniform prior over the 11 friction values of the grid, a GP discrepancy with the same priors and a posterior from the two draw-ins. M3 and M3n use histogram gradient boosting on geometry, force, lubrication, sheet thickness and oil film, with 500 incoming conditions drawn from calibration parts of the same pattern. Software: Python 3.11, SciPy and scikit-learn (Pedregosa et al, 2011).

5 Results

5.1 What the floor contains

Two batches of 50 consecutive parts of one alternative differ by up to 0.051 mm in mid-side draw-in and 0.078° in wall angle in 95 % of batch pairs (Table 4), so one floor is 0.7 to 1.7 long-term standard deviations of the parts. The floors are set by drift within the run (Fig. 4): without the time order they shrink by a factor of 2.1 to 4.5, and batches one position apart differ by 0.51–0.80 floors, nine apart by 1.1–1.5. They vary between runs: the floor of a single run is 0.21 to 2.16 times the pooled floor of its geometry, and the alternative-cluster interval is wider than the batch interval. Without the warm-up of the first 150 parts the floors fall to 0.75–1.00 of the reference, which keeps the warm-up because series production starts up after every break. Independent part-to-part noise, scanner repeatability included, accounts for 15–33 % of each floor, and the resolution is small against it except for the mid-side draw-in, so A3 holds for repeatability and resolution, while bias, linearity and stability remain untested.

Physical candidates for the drift are the warm-up of the tools and its effect on the lubricant, the build-up of lubricant on die and blank holder over consecutive strokes, and variation of the material and of the blank size along the stack, which enters the draw-in one to one because it is computed against the nominal 210 mm blank. Whether the drift is the process or the scanner cannot be settled without a reference artefact, but two checks bear on it. No recorded process signal explains it: batch centres of press force, punch temperature, sheet thickness and oil film predict it out of fold with R^2 of -0.23 to 0.20. And the batch centres of the flange width and of the cup width at 15 mm depth, which punch and die largely fix, are weakly correlated within the runs ($r = -0.27$ and $+0.27$), which argues against a common lateral scale drift of the scanner; the cup width varies by 0.06–0.07 mm between batches, so it bounds such a drift only below 0.05 %, and drift in height and of the edge detection remain untestable. Variation between runs is not in the floor: the three lubrication series of one geometry and force differ by 1.2 (waviness) to 9.8 floors (dome) at the 95th percentile of their centre differences, lubrication effects included, whereas within a run the 95th percentiles of two batches of 100 parts differ by 0.92–1.49 floors in 95 % of pairs.

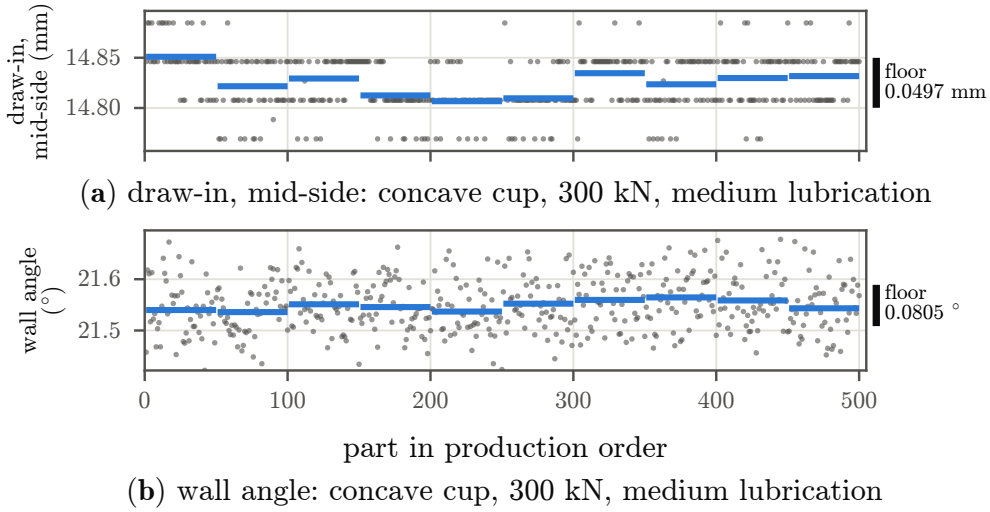
5.2 What production resolves and where the simulation dataset lies

All 63 geometry contrasts exceed the floor and 105 of 126 force contrasts do, but only 51 of 126 lubrication contrasts, so a lubrication variant is often a near-replicate of its siblings. The minimum resolvable force change is 21–67 kN between 100 and 300 kN for most characteristics and exceeds either force interval for the wall angle and the dome.

Table 4 Within-run floor per characteristic, pooled and per geometry, with its spread over runs

Characteristic	F	Interval	Concave/convex	Single runs	No warm-up	F/σ_{LT}
Mid-side (mm)	0.051	0.042–0.061	0.050/0.052	0.36–1.62	0.99	1.0/1.4
Corner (mm)	0.046	0.032–0.053	0.053/0.030	0.27–1.62	0.95	1.4/1.2
Wavin. (mm)	0.0028	0.0021–0.0050	0.0037/0.0021	0.21–2.08	0.89	1.6/0.7
Wall (°)	0.078	0.054–0.10	0.081/0.078	0.30–1.74	0.80	1.3/0.7
Arm (°)	0.11	0.089–0.15	0.12/0.100	0.40–2.16	1.00	1.7/0.8
Depth (mm)	0.011	0.0090–0.012	0.0087/0.012	0.39–1.45	0.75	1.4/1.6
Dome (mm)	0.0070	0.0052–0.0084	0.0034/0.0082	0.42–1.61	0.93	1.2/1.6

Batches of 50 consecutive parts, 95 % quantile of the difference of their 10 % trimmed means. Interval: 95 % over 2000 resamples of whole alternatives (the narrower interval over resampled batches is in the archive). Single runs: floor of one run over that of its geometry, range over the 18 runs. No warm-up: floor without the first 150 parts over the reference. σ_{LT}: standard deviation of the parts of a series (concave/convex); σ_{b}, σ_{w} and the drift ratio are in the archived results.



(**a**) draw-in, mid-side: concave cup, 300 kN, medium lubrication

(**b**) wall angle: concave cup, 300 kN, medium lubrication

Fig. 4 The 500 parts of one run in production order with the centres of its ten batches of 50 (bars) and the floor: mid-side draw-in (a) and wall angle (b)

The simulation dataset does not contain the produced process: at the nominal thickness the measured centre lies within the range spanned by the friction grid in only 11 of the 42 combinations of characteristic, geometry and force. The simulated mid-side draw-in exceeds the measured one at 100 kN for every friction coefficient (by 1.8–2.0 mm at friction 0.10), reaches it at 300 kN only at the edge of the grid on the concave cup, and at 500 kN at friction 0.105 and 0.116; the corner draw-in exceeds it everywhere, by 2.6–3.5 mm. The simulated force effect on the draw-in is 2.7 to 5.1 times the measured one, and the cut arms of the concave cups bend downwards in 89–100 % of the simulations but upwards in every part, whereas the convex arm angles agree within 0.6°. Calibrating friction cannot repair this within the grid: all 72 within-family least-squares calibrations of Mc end at its upper edge (0.15), and with a discrepancy term the posterior of Mcd piles at the lower edge (0.05), where the simulated force

effect is weakest. The dataset's mapping from oil film to friction implies 0.8–3.1 mm of mid-side draw-in per g/m^2 of oil, against within-run slopes of -0.01 to $+0.09$ mm and median slopes between the lubrication patterns of -0.17 to $+0.39$ mm per g/m^2; noise in the oil readings attenuates the within-run slope, but the mapping is too sensitive by an order of magnitude. An envelope over the measured incoming variation instead of the whole grid halves its width but makes M5 less decisive with a sibling (24 against 7.2 floors), because through this mapping it varies with the lubrication pattern more than production does. A Gaussian-process surrogate of the simulations decides as the simulation does (109 against 108 floors).

The cup depth is simulated 0.70–1.06 mm (2.5–3.7 %) deeper than measured, and the wall position at 15 mm depth, set by punch and die, 2.0–2.2 mm (2.8–3.5 %) smaller in radius. The depth offset is not definitional, since bottom and flange are seen from the same side of the sheet in both, nor due to the flange inclination, which would make the measured depth larger; a shorter effective drawing depth, a different tool geometry or a scanner scale error could each produce these offsets, which the data cannot separate. Without the depth offset, M0 decides at 65 instead of 108 floors (a sensitivity only).

5.3 Decisions by evidence relation

Table 5 and Fig. 5 score the rules; the nominal simulation decides at 108 floors, erring mostly by false rejects.

A new variant with a produced sibling. Every calibrated rule improves on it, and the production record is the most decisive: the nearest setting, the mean q_p of the two siblings, is right beyond 3.4 floors (2.0–4.5) and the force trend beyond 3.5, while M5 and M7n decide at 6.8–7.2 floors but are safe below one floor. Stratified by what a team knows when deciding, the nearest setting needs 1.60 floors when the two produced siblings do not differ resolvably from each other (75 cases on 17 alternatives) and 4.40 when they do (51 cases on 16), the force trend 3.05 and 4.20, and M5 6.15 and 8.90 (strata by the new design's own production, a diagnostic only: 0.85 and 4.85 floors). In physical units the nearest setting is right beyond 0.10 mm of mid-side draw-in, 0.26° of wall angle and 0.04 mm of cup depth (Table 6). One produced sibling suffices: with a single alternative at the new design's force the nearest setting decides at 3.5 floors (2.3–4.6), and further alternatives do not improve it, while they make M5 more decisive and safer.

A new process setting. With all alternatives at the new force held out, the force trend and the nearest setting decide at 8.2 and 8.5 floors and the interval rules only at 15–19, though they are safe at 5–8. Interpolating is much easier than extrapolating (the nearest setting 4.7 against 9.5 floors), but at 300 kN, the midpoint of a balanced calibration set, the nearest setting, the force trend and the centre of M5n coincide by construction, so interpolation cannot separate mean-type rules. The scaled simulation, extrapolating a fitted trend, falsely accepts failing alternatives up to 165 floors from their limit, on the cup depth, while on the mid-side draw-in and the arm angle it is more decisive than the force trend (4.0 against 5.7 and 5.9 against 9.5 floors extrapolating).

A new family. Scored in the floor of the produced family, as the procedure prescribes, every rule without the simulation fails (130–135 floors): nothing in one family's

Table 5 Decisive and safe distances (floors) by the relation of the new design to the produced evidence

	Sibling		Interpolated		Extrapolated		New family	
Rule	δ_d/δ_s	FA/FR	δ_d/δ_s	FA/FR	δ_d/δ_s	FA/FR	δ_d/δ_s	FA/FR
M0	108	17/109	109	17/115	106	12/109	104	28/105
M1s†	4.9	4.7/4.9	19	30/19	19	165/7.9	121	125/118
M2	23/<1	<1/<1	18/2.7	3.7/2.4	18/9.9	8.7/10.0	41/26	27/9.7
M5	7.2/<1	<1/<1	19/2.8	2.4/3.1	19/8.9	8.4/9.7	35/31	33/15
interval of δ_d	6.6–8.4		18–20		18–20		31–39	
M7n‡	6.8/<1	<1/<1	7.4/3.4	2.2/3.4	15/5.9	7.5/<1	132/130	129/167
interval of δ_d	6.0–7.3		7.1–8.1		15–16		131–132	
NN†	3.4	2.8/3.4	4.7	3.1/4.7	9.5	11/5.6	130	129/168
interval of δ_d	2.0–4.5		4.3–5.1		6.6–11		129–131	
M1sn†	3.5	3.5/3.6	4.7	3.1/4.7	8.9	9.3/7.4	131	129/168
NN+RM‡	6.3/<1	<1/<1	9.6/<1	<1/<1	20/<1	1.5/<1	293/4.5	4.1/44
Trial, first 20 parts	1.4	1.5/1.2	1.4	1.5/1.2	1.4	1.5/1.2	1.4	1.5/1.2
Trial, first 50 parts	0.90	1.0/0.85	0.90	1.0/0.85	0.90	1.0/0.85	0.90	1.0/0.85

Over all seven characteristics; one number for rules that always decide; <1: below one floor. FA/FR: safe distances on the false-accept and the false-reject side. 126 cases with a sibling and for a new family, 42 interpolated, 84 extrapolated; a new family in the produced family's floor (in its own floor in the archived results). Interval: 95 % over 1000 resamples of the held-out alternatives, fits fixed. Trial: the first n parts of the design's own run, a within-run lower bound. †, ‡: added after the first results and for this version.

Table 6 Decisive distance per characteristic in floors and, in brackets, in its unit

	NN			M1sn	M1s	M5	
Characteristic	Sibling	Interp.	Extrap.	Extrap.	Extrap.	New family	P_{pk}
Mid-side (mm)	2.0 (0.099)	3.8 (0.19)	6.5 (0.32)	5.7 (0.28)	4.0 (0.20)	5.5 (0.27)	1.3
Corner (mm)	4.5 (0.13)	1.9 (0.058)	6.7 (0.23)	2.8 (0.083)	2.9 (0.100)	9.9 (0.31)	2.5
Wavin. (mm)	<1 (<0.0028)	4.8 (0.014)	11 (0.040)	9.3 (0.027)	21 (0.077)	22 (0.047)	0.9
Wall (°)	3.4 (0.26)	<1 (<0.078)	1.9 (0.15)	1.8 (0.14)	2.0 (0.16)	7.5 (0.58)	1.7
Arm (°)	2.8 (0.28)	5.9 (0.59)	5.4 (0.53)	9.5 (0.94)	5.9 (0.69)	39 (3.9)	1.6
Depth (mm)	3.3 (0.039)	4.4 (0.038)	15 (0.13)	9.9 (0.086)	165 (1.4)	20 (0.20)	2.4
Dome (mm)	7.0 (0.038)	5.3 (0.044)	4.3 (0.020)	3.0 (0.020)	15 (0.056)	40 (0.21)	4.0

18 cases per characteristic with a sibling and for a new family, 6 interpolated, 12 extrapolated; units through the pooled floor of the characteristic; a new family in the produced family's floor. P_{pk}: median performance index of a requirement at the decisive distance of NN with a sibling. All rules and the safe distances are in the archived results.

record describes the other. With the simulation, Mcd decides at 30 floors and M5 at 35, and M3 is safe at 17 floors and M2 at 26. Per characteristic, M5 decides at 5–10 floors for the draw-ins and the wall angle, which the forming model represents directly, but at 20–40 for depth, waviness, arm and dome. Production knowledge still carries over where a drawing nominal transfers: for the wall angle, the design angle plus the produced family's deviation from its own decides at 1.9 floors, against 7.5 for M5. Two transfers do not resolve which rule is safest.

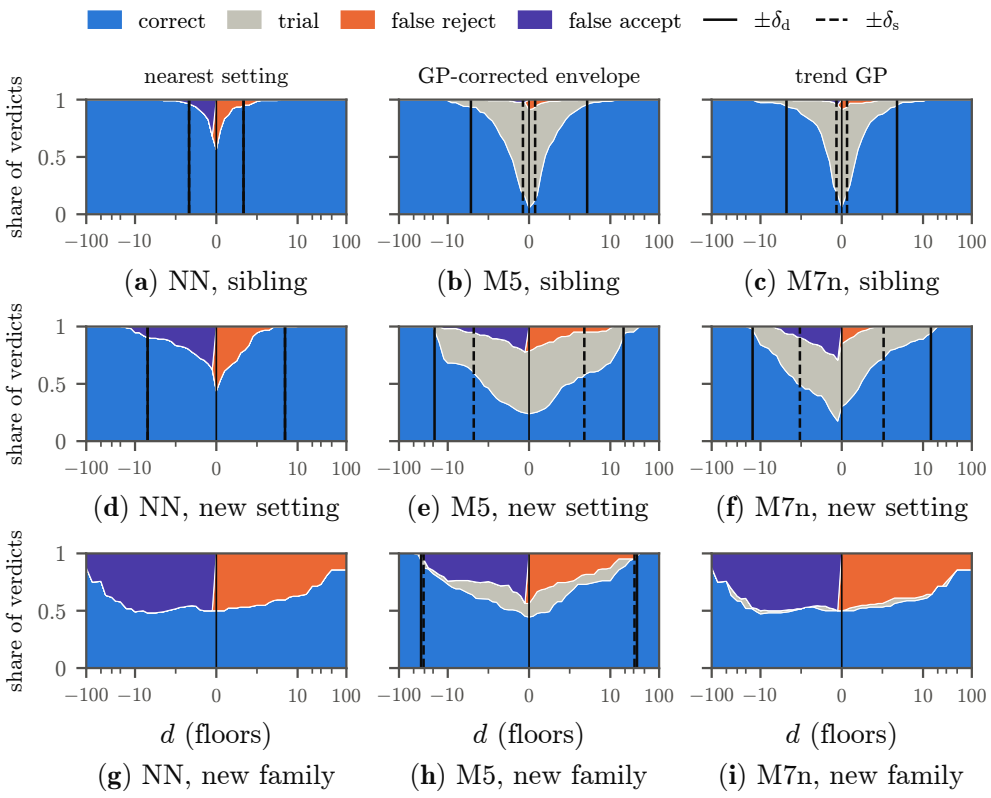


Fig. 5 Verdict shares against the requirement distance d (symmetric logarithmic axis, linear within ±10 floors; 126 cases per panel; a new family in the produced family's floor)

5.4 Learning given a fair test

The Gaussian processes fitted by maximum marginal likelihood reduce to the baselines in this design. For a new setting the force length scale of M5n sits at its lower bound in 68–79 % of the fits and its noise variance at its bound in every fit, and its centre lies within 0.01 floor of the calibration mean in 93–95 % of the cases; with a sibling it lies a median 0.28 floor from the nearest setting. The trend GP, specified for this regime, reduces to the force trend instead: its centre lies a median 0.33–0.36 floor from it for a new setting, its force length scale stays near the prior's 200 kN, which two force levels cannot inform, and its between-run nugget, 0.4–0.5 floors, is three to four times the sampling error of the truths. It is more decisive than M5n for a new setting (14.5 against 17.2 floors; paired difference −2.7, −5.6 to −1.1) and as safe, but less decisive than the nearest setting in every relation (by 3.5 floors with a sibling and 6 for a new setting). With the simulation in its mean (M7) it gets worse for a new setting (28 floors), because the simulated force trend enters it. The archive's study of part-level boosting (regularised, flexible, ridge, quantile loss, inner selection) finds no setting closer than 11.2 floors with a sibling and 13.1 for a new setting.

Table 7 Guard bands of step 6 (floors), their out-of-sample share correct and the trial rate

	Sibling				Interpolated				Extrapolated			
Rule	g	g_5	OOS	T	g	g_5	OOS	T	g	g_5	OOS	T
NN†	0.25	2.0	95	2	2.0	4.25	95	18	7.0	14.25	94	55
M1sn†	0.75	2.75	95	7	2.0	4.25	95	18	6.0	8.5	95	48
M2	0	0	99	54	0.75	2.75	95	53	none	none	–	100
M5	0	0	99	23	0.5	2.75	96	54	none	none	–	100
M7n‡	0	6.75	98	19	0.75	3.75	95	27	3.25	6.75	95	56
NN+RM‡	0	0.5	99	28	0	0	99	42	0	5.25	97	72

g, g_5: guard band under the reference prior and under the release-like prior (requirements within ±5 floors); none: no band up to 20 floors reaches 95 %. OOS: share correct (%) among decided verdicts with the band estimated leaving each alternative out within its family. T: share of verdicts sent to trial (%), reference prior. No rule qualifies for a new family; bands per family, out of sample and under other priors are in the archived results.

Prediction and interval separate. A relation-matched interval borrowed from the other family makes even the nearest setting safe for a new setting (safe distance <1 floor, against 6.8 for M5 and 5.2 for M7n) at the decisive distance of the learned intervals (19.5 against 18.7 and 14.5 floors), and covers 88 % of the truths interpolating and 93 % extrapolating, against 71 and 46 % for M5 and 52 and 56 % for M7n. By the interval score (Gneiting and Raftery, 2007) the relation-matched intervals are the best for a new setting (12–14 interpolating and 19–24 extrapolating, against 14–98 and 41–147 for the rules' own intervals). With a sibling the model-based intervals (M5, M5n, M7n, M7, Mcd) cover 83–86 % (cluster intervals 0.71–0.94), compatible with their nominal 90 %, and the jackknife+ intervals of M3 96 % (0.93–0.98) against a guarantee of 0.78; the conformal margin of M6, computed on sibling-type residuals, covers 93 % with a sibling at a cost of ten floors of decisiveness against M5 (17 against 7.2) but 61 % extrapolating. For a new family no interval covers well except at widths that send every design to trial. With one produced alternative of the new family, its offset to the other family's record decides at 8.9 floors in the new family's floor and the family's own record at 10.5; with two or three, its own nearest setting decides at 7.7 and 5.9 floors, while a two-task Gaussian process with a free inter-family correlation is safe at 1.3–1.6 floors but decides only at 13–17.

5.5 Guard bands

Step 5 turns the margins into guard bands (Table 7). Under the reference prior the nearest setting needs 0.25 floors with a sibling, 2.0 interpolating and 7.0 extrapolating, and sends 2, 18 and 55 % of the verdicts to trial; M2 and M5 need no band with a sibling and less than one floor interpolating, but send 23–54 % of the verdicts to trial, and they do not qualify for an extrapolated setting. Estimated per family, as step 5 prescribes, the bands differ between the geometries (the nearest setting 1.5 and 2.5 floors interpolating, 9.5 and 5.75 extrapolating), and M2 and M5 qualify for an extrapolated setting on the convex cup only; moved to the concave cup, their bands give only 84 and 83 % correct decided verdicts. Out of sample the bands hold: leaving one alternative out keeps the share of correct decided verdicts at 0.94–0.99 for every rule and relation, and leaving one force cell out at 0.93–0.95 for the nearest setting and

the force trend, but at 0.83 and 0.81 for M2 and M5 when extrapolating. The 95 % is an average over characteristics: with a sibling the nearest setting is right in 89 % of its decided verdicts on the dome, 98.5 % on the waviness and 92 % on the cut part.

The bands belong to the prior as much as to the rules. Under the release-like prior the band of the nearest setting grows to 2.0 floors with a sibling, 4.25 interpolating and 14.25 extrapolating, and it decides only 63, 32 and 2 % of the verdicts; leaving a force cell out, its share correct drops to 0.72 when extrapolating. Under the reference prior no rule qualifies for a new family, as the practice of trying out every new die would have it: Mcd reaches 95 % on one transfer with a band of 18 floors, but that band gives 73 % on the other (under a uniform prior within ±50 floors M2, M3 and M5 qualify). The applicability guards, reported as diagnostics, refuse indiscriminately: the range guard refuses exactly the extrapolated designs, and 96 % of its refusals at requirements 10 floors from the truth would have been right for the nearest setting.

5.6 Costs, trials and release-level requirements

Pricing the verdicts by Eq. (2) with a false reject at half the cost of a false accept (Fig. 6), the production record and the Gaussian processes cost about the same with a sibling while trials are cheap, and the nearest setting is the cheapest rule once a trial costs a tenth of a false accept or more; for a new setting the force trend or the nearest setting is the cheapest at every trial cost; and for a new family trying every design is the cheapest until a trial costs about a tenth of a false accept, beyond which M2 and then M5 are. The cost-minimising band of the nearest setting falls from 5 to 0 floors with a sibling and from 15 to 0 when extrapolating as the trial cost rises from 0.01 to 0.5 of a false accept; the 95 % bands of step 5 correspond to trials costing a tenth to a quarter of a false accept, more than a new force on an existing tool costs, so cost-optimal bands would be wider. A trial is not error-free either: scored as a rule, a trial of the first 20 or 50 parts of the design's own run decides at 1.35 and 0.90 floors, a lower bound, since a trial on another day also carries the variation between runs.

Upper limits at which an alternative has a performance index of 1.0, 1.33, 1.67 or 2.0 lie a median 1.03, 1.75, 2.46 and 3.14 floors above its 95th percentile. A limit at an index of 1.33, a recommended minimum for existing processes and the lower end of the range in which an initial process study may be acceptable after a customer review, therefore lies inside every attainable distance except that of the nearest setting when the produced siblings do not differ resolvably from each other (1.60 floors), which it exceeds by only 0.15 floor: release-level requirements are decided at the design stage only with a near-replicate produced sibling, if at all.

5.7 Sensitivity, selection and robustness

A between-run component added to every truth lengthens the distances but keeps their order: with $\tau = 3$ floors, about the largest variation between series, the nearest setting needs 5.8 floors with a sibling and 10 extrapolating, M5 9.3 and 20, a production-record rule (nearest setting, force trend or median) is the most decisive within the family in each of 200 draws, and the band of the nearest setting with a sibling grows from 0.25 to 2.9 floors. Selected on one geometry and evaluated on the other, the most decisive

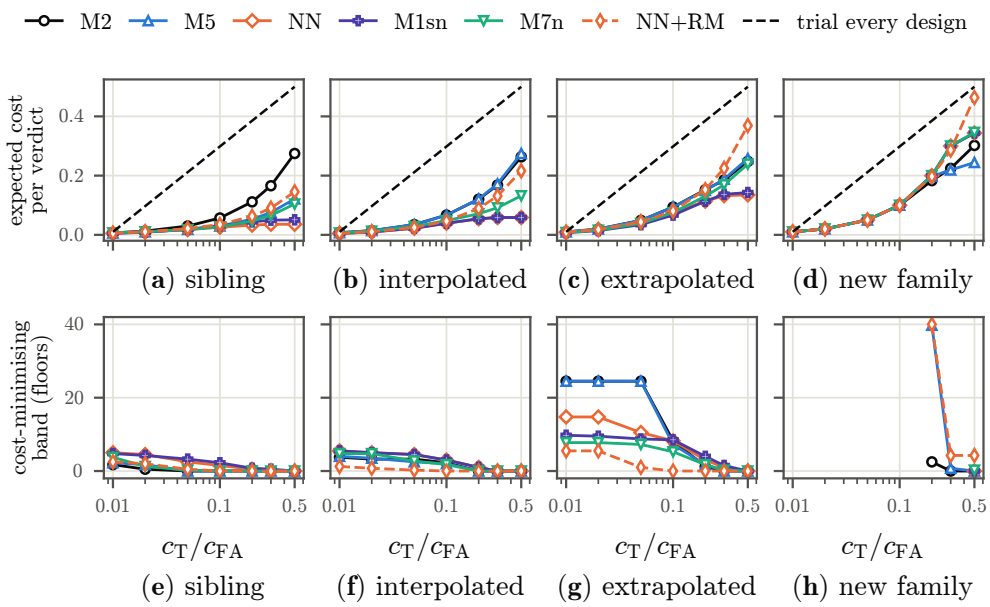


Fig. 6 Expected cost per verdict with the cost-minimising band (a–d) and that band (e–h) against the cost of a trial; reference prior, $c_{FR} = 0.5\,c_{FA}$

rule is the nearest setting with a sibling (2.0 and 3.6 floors), the force trend or the median interpolating (4.2–5.8) and the force trend or the nearest setting extrapolating (8.9–11); for a new family the rule selected on one transfer decides at 32–33 floors on the other, and the safest pre-specified rule, M3, is safe at 17–18. Paired intervals between point rules span about ±2 floors with a sibling and when extrapolating, so smaller differences are not resolved by this design.

Requiring all seven characteristics of a design to meet raises the decisive distances of the production-record rules by at most 1.8 floors within a family (the nearest setting 5.1 floors with a sibling, the force trend 9.5 extrapolating) but those of M5 by up to 16, and to 39 floors for M5 for a new family. Lower limits on the 5th percentile, calibration and truths from the first 20 or 50 parts, and the decided statistic $\bar{m}+4\sigma$ (an index of 1.33) leave the nearest setting within 0.4 floor of its reference distance with a sibling and within 1.2 when extrapolating, and a production-record rule the most decisive within the family, as do variants of data, constants and specifications; other floor protocols rescale the distances by 0.60 to 1.42, and floors recomputed without the held-out alternative's own series move the distances of the nearest setting and the force trend by at most 0.25 floors and those of the other main rules by at most 1.4.

6 Discussion

6.1 What the evaluation establishes

Two findings follow from the definitions and hold wherever the framework is applied: the floor puts every rule and characteristic on a common production-referenced scale,

Table 8 Guidance for design-stage manufacturability decisions, with its basis

Situation	Guidance	Basis
From the framework (prescriptive; usefulness not evaluated)		
Before relying on a predictor	Measure the floor on consecutive production of the family with a stated protocol; report σ_b, σ_w, its lag dependence and the floors of single runs; treat it as a lower bound of series variation.	framework
Qualifying a rule	Evaluate it by grouped cross-validation for the evidence relation of the coming designs, per family, with siblings stratified by whether they differ from each other, and per characteristic.	framework
Choosing a rule	Compare it with the nearest produced setting and with its own version without the simulation, on both distances and by expected cost at the team's trial cost.	framework
Deciding a single design	Derive the guard band for the requirements the team expects, validate it out of sample, and apply it to the observable margin; a margin inside it means a trial.	framework
Observed in this study (one process, 18 alternatives produced once, 2 geometries; not general)		
New variant with a produced sibling	the nearest produced setting decided beyond 3.4 floors (1.6 when the produced siblings did not differ resolvably from each other); no learned rule was more decisive	18 alternatives
New process setting	production record 8.2–8.5 floors, interval rules 15–19; intervals matched to the relation made it safe	6 force cells
New family	only rules with the (generic) simulation decided, at 30 floors and beyond; none qualified for single verdicts	two transfers
Release-level requirements	within about 2 floors of the 95th percentile: decided at best with a near-replicate sibling, otherwise by trial	18 alternatives

though not on a common scale of nonconforming fractions, and as a within-run quantity it bounds the variation of series production from below (RQ1); and an interval rule trades decisiveness for safety, which the two distances make explicit. The others are observations on one laboratory process, with every alternative produced once and a generic simulation dataset, and hypotheses for confirmation. The attainable distance grows with the novelty of the design: a production-record rule committed to in advance decided at 3.4 floors with a sibling (1.6 when the two produced siblings did not differ resolvably, 4.4 when they did), 8.5 for a new setting (4.7 interpolating, 9.5 extrapolating) and 30 floors at best for a new family, and within a family the choice of rule mattered more than the relation, also among the pre-specified rules (RQ2). For this simulation dataset and for Gaussian processes and boosting fitted to six to nine alternatives, the simulation contributed little within a family, helping single characteristics when extrapolating in force, and mattered across families, for the characteristics it models directly; the learners added no decisiveness, and what made rules safe for new settings was an interval calibrated on the relation (RQ3). Qualification by relation also exposed a dangerous extrapolation: the scaled simulation, decisive with a sibling, accepted failing alternatives up to 165 floors from their limit when extrapolating.

6.2 Implications for practice

For a process planner or quality engineer the results say three things. Within a family the production record answers most design-stage questions if it is indexed by how the

coming design relates to it: a sibling at the same setting settles a requirement a few floors away, a new force within the produced range almost as well, and an extrapolated force or a new family needs a trial unless the requirement is far. Under the reference prior, of 100 verdicts on new variants with a sibling the nearest setting with its band sends 2 to trial and gets at least 93 of the other 98 right; of 100 on extrapolated settings it sends 55 to trial. And near release-level limits nothing replaced a trial without a nearly identical produced sibling, so the value of design-stage prediction lies in the many decisions far from the limits rather than in the few close ones. For a developer of an AI-based manufacturability tool the lesson is methodological: a learned model has to be compared with a production-record baseline and with its own version without the simulation, scored by evidence relation in a unit that production defines, its intervals calibrated on cases of the same relation and shown to its user with a guard band. The nearest setting is also the most transparent rule, pointing to the produced alternatives whose record it averages, and so the natural default where a learned rule shows no advantage.

6.3 Simulation and machine learning in this regime

Within a design family the production record carried the decision even against a calibrated or rescaled simulation, but the simulation here is a generic dataset that does not contain the produced process: its force trend is two to five times too large, its friction calibration is censored at an edge of the grid with or without a discrepancy, and its oil-to-friction mapping is too sensitive. A process-calibrated simulation, with friction depending on pressure (Hol et al, 2012), a hardening model fit for springback (Wagoner et al, 2013) and elastic tools (Pilthammar et al, 2018), might contribute more; the per-characteristic results show where it already helps, for the draw-ins and the arm angle when extrapolating and for the characteristics it models directly in a new family. The learners failed for a reason the framework exposes: with six to nine alternatives at two or three force levels a flexible model can identify at most a linear trend, and a well-specified one then becomes that trend. Nonlinear multi-fidelity models (Perdikaris et al, 2017), physics-informed learning (Karniadakis et al, 2021), conformalised quantile regression on the parts (Romano et al, 2019) and multi-task models over more families (Bonilla et al, 2008) are the next candidates, to be scored in the same way.

6.4 Reuse for other processes and AI-based tools: a proposal

The framework needs several produced variants of a design family whose relations span those of the coming designs, a series of consecutive parts or replicated batches per variant, a design-stage predictor for every variant and characteristics measured identically on parts and predictions; here nine produced alternatives per family and 500-part series gave ten batches of 50 parts for the floor, one produced sibling sufficed for the nearest setting, and truths from the first 20 or 50 parts moved its distance with a sibling by less than half a floor. The extensions below are untested. A classifier with a fixed threshold does not take the requirement as input, so its operating characteristic is estimated by binning test cases by their true margin, which needs a continuous,

Table 9 Proposed reporting checklist for evaluating a design-stage decision rule against production

Item	Why it matters	In this paper
Floor and its protocol	every distance is in this unit	B, α, lag, single runs (Table 4); σ_b, σ_w archived
Measurement of the characteristics	biases enter absolute comparisons	Table 3, Section 4.3
Relations, cases and replication units	alternatives, not cases, replicate	per relation, 18 alternatives
Both distances, false-accept side	decisiveness and safety differ	Table 5
Baselines	production record, no simulation	NN, M1sn; every pair with and without
Intervals: source and level	coverage is a property of the construction	Table 2, Section 5.4
Per-characteristic results	pooled distances hide the worst	Table 6
Guard band, prior, validation	the deployed part	per family, out of sample (Table 7)
Costs and trial rates	deployment is a cost question	Fig. 6
Development record	selection optimism	Section 4.5, cross-fitting
Query protocol (language models)	verdicts vary between queries	proposal (Section 6.4)

production-referenced outcome for every test design that CAD-based manufacturability datasets usually lack. A language-model check is scored on a grid of requirements with repeated queries, reporting model and version, query date, decoding settings and prompt paraphrases, treating refusals and hedged answers as trials, checking that its verdicts are monotone in the requirement and testing for contamination, since public datasets and the papers describing them may be in its training data. A requirement on a rate, such as the share of split or wrinkled parts, is scored in the same way once each alternative's rate is estimated from its series, with a floor over batches of rates; for continuous design spaces the relation becomes a distance from the produced designs in descriptor space. Table 9 is a proposed reporting checklist, modelled on model cards (Mitchell et al, 2019) and TRIPOD+AI (Collins et al, 2024); the scoring is released as a small documented function, separate from the pipeline, that takes intervals, truths and floors and returns distances, guard bands and trial rates (Canseven, 2026). An attempt on open powder-bed fusion data (Leirmo, 2021) failed for lack of replicated batches within a build: a floor within a build would need at least ten replicated anchors of one design per build and several builds per setting.

6.5 Deployment

In use, a step-6 verdict would be shown with its margin in floors and in the unit of the characteristic, the relation on which it rests and the produced alternatives behind it, and a trial verdict with the requirement distance at which the rule would decide. A qualification has to be maintained: the realised errors of step-6 verdicts can be tracked against later production on a control chart of decision outcomes, and a rule re-qualified when its share of wrong decided verdicts exceeds 5 % beyond what the

number of verdicts explains, when the floor of new runs leaves the range of the single-run floors, or when the die is reworked. The evidence has a cost, several 500-part series per family before any rule qualifies, which the results bound from below: one produced sibling settles most verdicts with a sibling, and one produced alternative of a new family brings its decisive distance from 28 to 9 floors in its own floor. Whether teams decide better with the framework is an empirical question: a study with cases of known production outcome could test whether designers shown the guard-banded verdict accept fewer wrong verdicts within the safe distance than designers shown the verdict alone, at an equal rate of right acceptances (Lee and See, 2004), and whether the band narrows the spread of that share between teams (Zhang et al, 2021).

6.6 Threats to validity

The evaluation covers one material, one press and two geometries, with relations within a family from a 3×3 factorial in which lubrication is mostly not resolved; the procedure is general, the numbers are not. The new family rests on two transfers between geometries that differ in three features at once. Each alternative was produced once, so the reference q_p is that of one run and the variation between days, coils and tool states is missing; the sensitivity analysis bounds its effect but cannot replace replicated runs. The floor does not separate scanner drift from process drift, and absolute values are uncalibrated. The simulation dataset was not calibrated to the process, so the findings on simulation concern this dataset. The requirements are one-sided limits on the 95th percentile of dimensional characteristics, below the conformance of series release; attribute outcomes such as splits and wrinkles are not covered; and the verdicts are marginal per characteristic, a design-level verdict being an extension. The guard band was estimated pooled over characteristics, its prior is a reference rather than a team's, and the costs are assumed. Rules, strata, bands and priors were partly chosen after results had been seen and are evaluated on the same data; the cross-fitting bounds the optimism but does not remove it, and a single analyst carried out the analysis, with code and data open for replication.

7 Conclusions

Design-stage manufacturability decisions can be evaluated against production on a scale production defines: the decisive and safe distances in floors tell how close to a requirement a rule decides and how close it can be relied on not to mislead, and a guard band validated for the requirements a team expects turns them into a rule for single decisions, priced by the costs of errors and trials. In an exploratory application to one laboratory deep-drawing process, with alternatives produced once and a generic simulation dataset, the attainable distance grew with the novelty of the design: the nearest produced setting, committed to in advance, decided at 3.4 floors with a produced sibling and 8.5 for a new process setting, and no rule decided closer than 30 floors for a new family (the pre-specified M5 at 35). Release-level requirements lay within these distances, so they needed a trial unless a near-replicate sibling was produced.

For design practice, the production record carried the decision within a family, which argues for indexing such records by their relation to the coming design rather

than for a more elaborate model; across families only rules using the simulation carried the change of geometry. For AI-based manufacturability support, the evaluation separated what the learners were credited with: Gaussian processes and boosting fitted to six to nine alternatives reproduced the production record, and intervals calibrated on the same evidence relation, not learning, made it safe for new settings.

The evidence is one process and two geometries, and the comparison of rules is exploratory. Four extensions follow: series from several days, coils and tool states; a process-calibrated simulation; families with continuous geometric descriptors; and a confirmatory test of the frozen procedure on other data. Within a design family, the baseline that a learned manufacturability check has to beat is a production record indexed by its relation to the coming design.

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Data availability. All data are public. RDDAC: doi:10.18419/DARUS-5589 (version 2.0); DDACS: doi:10.18419/DARUS-4801 (version 3.0); both licensed CC BY 4.0. The extracted per-part and per-simulation feature tables and all result files are archived with the code (Canseven, 2026).

Code availability. The complete analysis pipeline, which reproduces every number in this paper from the public data, is archived on Zenodo (Canseven, 2026); `results/findings.md` maps every number to the script and result file that produce it, and `scripts/opchar.py` is the scoring function for other datasets; M3n is named M4 in the code. A working copy is at https://github.com/htcanseven/Claude.

Author contributions. H.T.C. conceived the method, designed and carried out the analysis, and wrote the paper.

Use of generative AI. A large language model (Claude, Anthropic) was used under the author's direction to write and debug code, retrieve and check literature, run analyses and draft text. It was also used to simulate reviews of earlier versions of the manuscript in five rounds, acting as referees and editor; most rules and analyses added after the first results, listed in Section 4.5, were added in response to these simulated reviews. None of these reviews is a review by a journal. The author designed the study, verified every analysis against the archived results, checked every reference against its published record, and takes full responsibility for the content; the language model is not an author. All figures come from the released code; none is AI-generated.

References

Arendt PD, Apley DW, Chen W (2012) Quantification of model uncertainty: Calibration, model discrepancy, and identifiability. Journal of Mechanical Design 134(10):100908. https://doi.org/10.1115/1.4007390

ASME (2018) Assessing credibility of computational modeling through verification and validation: Application to medical devices. ASME V&V 40-2018, American Society of Mechanical Engineers, New York

Automotive Industry Action Group (2006) Production Part Approval Process (PPAP), 4th edn. AIAG, Southfield, MI

Automotive Industry Action Group (2010) Measurement Systems Analysis, 4th edn. AIAG, Southfield, MI

Ballegeer M, Benoit DF (2026) BenDFM: A taxonomy and synthetic CAD dataset for manufacturability assessment in sheet metal bending. Journal of Intelligent Manufacturing https://doi.org/10.1007/s10845-026-02857-9

Banabic D (2010) Sheet Metal Forming Processes: Constitutive Modelling and Numerical Simulation. Springer, Berlin, Heidelberg, https://doi.org/10.1007/978-3-540-88113-1

Barber RF, Candès EJ, Ramdas A, et al (2023) Conformal prediction beyond exchangeability. The Annals of Statistics 51(2):816–845. https://doi.org/10.1214/23-aos2276

Bates S, Hastie T, Tibshirani R (2024) Cross-validation: What does it estimate and how well does it do it? Journal of the American Statistical Association 119(546):1434–1445. https://doi.org/10.1080/01621459.2023.2197686

Baum S, Heinzelmann P (2025) Deep drawing and cutting simulations (DDACS) dataset. DaRUS, University of Stuttgart, version 3.0, https://doi.org/10.18419/DARUS-4801

Baum S, Heinzelmann P (2026) Real deep drawing and cutting (RDDAC) dataset. DaRUS, University of Stuttgart, version 2.0, https://doi.org/10.18419/DARUS-5589

Baum S, Heinzelmann P, Clauß P, et al (2026a) Statistical analysis of simulation-to-reality deviation in deep drawing with a benchmark dataset. Transactions of the Indian Institute of Metals 79(7):176. https://doi.org/10.1007/s12666-026-03870-5

Baum S, Heinzelmann P, Liewald M, et al (2026b) Addressing discrepancies between synthetic and real-world data of deep-drawing manufacturing systems in AI applications. Procedia CIRP 138:727–732. https://doi.org/10.1016/j.procir.2026.01.125

Bechhofer RE (1954) A single-sample multiple decision procedure for ranking means of normal populations with known variances. The Annals of Mathematical Statistics 25(1):16–39. https://doi.org/10.1214/aoms/1177728845

Bonilla EV, Chai KMA, Williams CKI (2008) Multi-task Gaussian process prediction. In: Advances in Neural Information Processing Systems, vol 20. Curran Associates, Inc., pp 153–160

Boothroyd G, Dewhurst P, Knight WA (2010) Product Design for Manufacture and Assembly, 3rd edn. CRC Press, Boca Raton, https://doi.org/10.1201/9781420089288

Brynjarsdóttir J, O'Hagan A (2014) Learning about physical parameters: the importance of model discrepancy. Inverse Problems 30(11):114007. https://doi.org/10.1088/0266-5611/30/11/114007

Canseven HT (2026) Code, feature tables and results for "when can a prediction replace a physical trial? A production-referenced evaluation of design-stage decision rules in deep drawing". Zenodo, version 2, *[DOI pending]*

Cao J, Bambach M, Merklein M, et al (2024) Artificial intelligence in metal forming. CIRP Annals 73(2):561–587. https://doi.org/10.1016/j.cirp.2024.04.102

Cawley GC, Talbot NLC (2010) On over-fitting in model selection and subsequent selection bias in performance evaluation. Journal of Machine Learning Research 11:2079–2107. URL https://jmlr.org/papers/v11/cawley10a.html

Chen K, Breunig A, Ha J, et al (2022) Robustness of deep-drawing finite-element simulations to process variations. International Journal of Material Forming 15(3):45. https://doi.org/10.1007/s12289-022-01695-3

Cheng FT, Chen YT, Su YC, et al (2008) Evaluating reliance level of a virtual metrology system. IEEE Transactions on Semiconductor Manufacturing 21(1):92–103. https://doi.org/10.1109/tsm.2007.914373

Cheng FT, Huang HC, Kao CA (2012) Developing an automatic virtual metrology system. IEEE Transactions on Automation Science and Engineering 9(1):181–188. https://doi.org/10.1109/tase.2011.2169405

Chow CK (1970) On optimum recognition error and reject tradeoff. IEEE Transactions on Information Theory 16(1):41–46. https://doi.org/10.1109/TIT.1970.1054406

Collins GS, Moons KGM, Dhiman P, et al (2024) TRIPOD+AI statement: updated guidance for reporting clinical prediction models that use regression or machine learning methods. BMJ 385:e078378. https://doi.org/10.1136/bmj-2023-078378

Dreyfus PA, Psarommatis F, May G, et al (2022) Virtual metrology as an approach for product quality estimation in Industry 4.0: a systematic review and integrative conceptual framework. International Journal of Production Research 60(2):742–765. https://doi.org/10.1080/00207543.2021.1976433

Eckert C, Isaksson O, Earl C (2019) Design margins: a hidden issue in industry. Design Science 5:e9. https://doi.org/10.1017/dsj.2019.7

Elkan C (2001) The foundations of cost-sensitive learning. In: Proceedings of the Seventeenth International Joint Conference on Artificial Intelligence (IJCAI 2001). Morgan Kaufmann, pp 973–978

Fischer P, Harsch D, Heingärtner J, et al (2018) Implementation of feedback control in kitchen sink production. IOP Conference Series: Materials Science and Engineering 418:012110. https://doi.org/10.1088/1757-899x/418/1/012110

Geifman Y, El-Yaniv R (2017) Selective classification for deep neural networks. In: Advances in Neural Information Processing Systems, vol 30. Curran Associates, Inc., pp 4878–4887

Ghadai S, Balu A, Sarkar S, et al (2018) Learning localized features in 3D CAD models for manufacturability analysis of drilled holes. Computer Aided Geometric Design 62:263–275. https://doi.org/10.1016/j.cagd.2018.03.024

Gneiting T, Raftery AE (2007) Strictly proper scoring rules, prediction, and estimation. Journal of the American Statistical Association 102(477):359–378. https://doi.org/10.1198/016214506000001437

Havinga J, Mandal PK, van den Boogaard T (2020) Exploiting data in smart factories: real-time state estimation and model improvement in metal forming mass production. International Journal of Material Forming 13(5):663–673. https://doi.org/10.1007/s12289-019-01495-2

Hazelrigg GA (1998) A framework for decision-based engineering design. Journal of Mechanical Design 120(4):653–658. https://doi.org/10.1115/1.2829328

Heddoub A, Diallo AR, Homri L, et al (2026) Conformal prediction for multivariate quality assurance – application on coordinate measuring machines. Procedia CIRP 145:91–96. https://doi.org/10.1016/j.procir.2026.03.151

Heinzelmann P, Baum S, Riedmüller KR, et al (2025a) A comprehensive benchmark dataset for sheet metal forming: Advancing machine learning and surrogate modelling in process simulations. MATEC Web of Conferences 408:01090. https://doi.org/10.1051/matecconf/202540801090

Heinzelmann P, Baum S, Riedmüller KR, et al (2025b) A data-driven surrogate model for predicting springback in deep drawing processes. Journal of Physics: Conference Series 3104(1):012060. https://doi.org/10.1088/1742-6596/3104/1/012060

Higdon D, Kennedy M, Cavendish JC, et al (2004) Combining field data and computer simulations for calibration and prediction. SIAM Journal on Scientific Computing 26(2):448–466. https://doi.org/10.1137/s1064827503426693

Hol J, Cid Alfaro MV, de Rooij MB, et al (2012) Advanced friction modeling for sheet metal forming. Wear 286–287:66–78. https://doi.org/10.1016/j.wear.2011.04.004

ISO (1994) Accuracy (trueness and precision) of measurement methods and results – part 6: Use in practice of accuracy values. ISO 5725-6:1994, International Organization for Standardization, Geneva

ISO (2017) Geometrical product specifications (GPS) – inspection by measurement of workpieces and measuring equipment – part 1: Decision rules for verifying conformity or nonconformity with specifications. ISO 14253-1:2017, International Organization for Standardization, Geneva

ISO (2023) Accuracy (trueness and precision) of measurement methods and results – part 3: Intermediate precision and alternative designs for collaborative studies. ISO 5725-3:2023, International Organization for Standardization, Geneva

ISO (2025) Accuracy (trueness and precision) of measurement methods and results – part 2: Basic method for the determination of repeatability and reproducibility of a standard measurement method. ISO 5725-2:2025, International Organization for Standardization, Geneva

ISO (2026) Statistical methods in process management – capability and performance – part 2: Process capability and performance of time-dependent process models. ISO 22514-2:2026, International Organization for Standardization, Geneva

Joint Committee for Guides in Metrology (2012) Evaluation of measurement data – the role of measurement uncertainty in conformity assessment. JCGM 106:2012, BIPM, Sèvres, https://doi.org/10.59161/jcgm106-2012

Karniadakis GE, Kevrekidis IG, Lu L, et al (2021) Physics-informed machine learning. Nature Reviews Physics 3(6):422–440. https://doi.org/10.1038/s42254-021-00314-5

Kennedy MC, O'Hagan A (2000) Predicting the output from a complex computer code when fast approximations are available. Biometrika 87(1):1–13. https://doi.org/10.1093/biomet/87.1.1

Kennedy MC, O'Hagan A (2001) Bayesian calibration of computer models. Journal of the Royal Statistical Society Series B: Statistical Methodology 63(3):425–464. https://doi.org/10.1111/1467-9868.00294

Le Gratiet L, Garnier J (2014) Recursive co-kriging model for design of computer experiments with multiple levels of fidelity. International Journal for Uncertainty Quantification 4(5):365–386. https://doi.org/10.1615/int.j.uncertaintyquantification.2014006914

Lee JD, See KA (2004) Trust in automation: Designing for appropriate reliance. Human Factors: The Journal of the Human Factors and Ergonomics Society 46(1):50–80. https://doi.org/10.1518/hfes.46.1.50.30392

Lei J, G'Sell M, Rinaldo A, et al (2018) Distribution-free predictive inference for regression. Journal of the American Statistical Association 113(523):1094–1111. https://doi.org/10.1080/01621459.2017.1307116

Leirmo TL (2021) Data on dimensional and geometric accuracy for laser-based powder bed fusion of PA12 (PA2200). DataverseNO, version 1.1, https://doi.org/10.18710/DHACHZ

Madras D, Pitassi T, Zemel R (2018) Predict responsibly: Improving fairness and accuracy by learning to defer. In: Advances in Neural Information Processing Systems, vol 31. Curran Associates, Inc., pp 6147–6157

Makatura L, Foshey M, Wang B, et al (2024) How can large language models help humans in design and manufacturing? Part 1: Elements of the LLM-enabled computational design and manufacturing pipeline. Harvard Data Science Review (Special Issue 5). https://doi.org/10.1162/99608f92.cc80fe30

Malak RJ, Paredis CJJ (2010) Using support vector machines to formalize the valid input domain of predictive models in systems design problems. Journal of Mechanical Design 132(10):101001. https://doi.org/10.1115/1.4002151

Maschler B, Weyrich M (2021) Deep transfer learning for industrial automation: A review and discussion of new techniques for data-driven machine learning. IEEE Industrial Electronics Magazine 15(2):65–75. https://doi.org/10.1109/mie.2020.3034884

Mitchell M, Wu S, Zaldivar A, et al (2019) Model cards for model reporting. In: Proceedings of the Conference on Fairness, Accountability, and Transparency (FAT* '19). Association for Computing Machinery, pp 220–229, https://doi.org/10.1145/3287560.3287596

Montgomery DC (2019) Introduction to Statistical Quality Control, 8th edn. Wiley, Hoboken, NJ

Nemani V, Biggio L, Huan X, et al (2023) Uncertainty quantification in machine learning for engineering design and health prognostics: A tutorial. Mechanical Systems and Signal Processing 205:110796. https://doi.org/10.1016/j.ymssp.2023.110796

Oberkampf WL, Roy CJ (2010) Verification and Validation in Scientific Computing. Cambridge University Press, Cambridge, https://doi.org/10.1017/cbo9780511760396

Pahl G, Beitz W, Feldhusen J, et al (2007) Engineering Design: A Systematic Approach, 3rd edn. Springer, London, https://doi.org/10.1007/978-1-84628-319-2

Pedregosa F, Varoquaux G, Gramfort A, et al (2011) Scikit-learn: Machine learning in Python. Journal of Machine Learning Research 12(85):2825–2830. URL https://jmlr.org/papers/v12/pedregosa11a.html

Perdikaris P, Raissi M, Damianou A, et al (2017) Nonlinear information fusion algorithms for data-efficient multi-fidelity modelling. Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences 473(2198):20160751. https://doi.org/10.1098/rspa.2016.0751

Picard C, Edwards KM, Doris AC, et al (2025) From concept to manufacturing: evaluating vision-language models for engineering design. Artificial Intelligence Review 58(9):288. https://doi.org/10.1007/s10462-025-11290-y

Pilthammar J, Sigvant M, Kao-Walter S (2018) Introduction of elastic die deformations in sheet metal forming simulations. International Journal of Solids and Structures 151:76–90. https://doi.org/10.1016/j.ijsolstr.2017.05.009

Rasmussen CE, Williams CKI (2006) Gaussian Processes for Machine Learning. MIT Press, Cambridge, MA, https://doi.org/10.7551/mitpress/3206.001.0001

Roberts DR, Bahn V, Ciuti S, et al (2017) Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. Ecography 40(8):913–929. https://doi.org/10.1111/ecog.02881

Romano Y, Patterson E, Candès EJ (2019) Conformalized quantile regression. In: Advances in Neural Information Processing Systems, vol 32. Curran Associates, Inc., pp 3538–3548

Salado A, Kannan H (2018) A mathematical model of verification strategies. Systems Engineering 21(6):593–608. https://doi.org/10.1002/sys.21463

Tercan H, Meisen T (2022) Machine learning and deep learning based predictive quality in manufacturing: a systematic review. Journal of Intelligent Manufacturing 33(7):1879–1905. https://doi.org/10.1007/s10845-022-01963-8

Tercan H, Guajardo A, Heinisch J, et al (2018) Transfer-learning: Bridging the gap between real and simulation data for machine learning in injection molding. Procedia CIRP 72:185–190. https://doi.org/10.1016/j.procir.2018.03.087

Thomke S, Bell DE (2001) Sequential testing in product development. Management Science 47(2):308–323. https://doi.org/10.1287/mnsc.47.2.308.9838

Varma S, Simon R (2006) Bias in error estimation when using cross-validation for model selection. BMC Bioinformatics 7:91. https://doi.org/10.1186/1471-2105-7-91

Vickers AJ, Elkin EB (2006) Decision curve analysis: A novel method for evaluating prediction models. Medical Decision Making 26(6):565–574. https://doi.org/10.1177/0272989x06295361

Vovk V, Gammerman A, Shafer G (2005) Algorithmic Learning in a Random World. Springer, New York, https://doi.org/10.1007/b106715

Wagoner RH, Lim H, Lee MG (2013) Advanced issues in springback. International Journal of Plasticity 45:3–20. https://doi.org/10.1016/j.ijplas.2012.08.006

Whitney DE (1996) Why mechanical design cannot be like VLSI design. Research in Engineering Design 8(3):125–138. https://doi.org/10.1007/bf01608348

Wiebenga JH, Atzema EH, An YG, et al (2014) Effect of material scatter on the plastic behavior and stretchability in sheet metal forming. Journal of Materials Processing Technology 214(2):238–252. https://doi.org/10.1016/j.jmatprotec.2013.08.008

Wuest T, Weimer D, Irgens C, et al (2016) Machine learning in manufacturing: advantages, challenges, and applications. Production & Manufacturing Research 4(1):23–45. https://doi.org/10.1080/21693277.2016.1192517

Yan X, Wang Z, Zhao C, et al (2026) Machine learning-based cyber manufacturing services: A review of manufacturing process selection, process planning, and design for manufacturing. Engineering 63:47–67. https://doi.org/10.1016/j.eng.2025.11.034

Yoshida F, Uemori T (2002) A model of large-strain cyclic plasticity describing the Bauschinger effect and workhardening stagnation. International Journal of Plasticity 18(5–6):661–686. https://doi.org/10.1016/S0749-6419(01)00050-X

Zaoui A, Denis C, Hebiri M (2020) Regression with reject option and application to kNN. In: Advances in Neural Information Processing Systems, vol 33. Curran Associates, Inc., pp 20073–20082

Zhang G, Raina A, Cagan J, et al (2021) A cautionary tale about the impact of AI on human design teams. Design Studies 72:100990. https://doi.org/10.1016/j.destud.2021.100990

Zhang Y, Liao QV, Bellamy RKE (2020) Effect of confidence and explanation on accuracy and trust calibration in AI-assisted decision making. In: Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency (FAT* '20). Association for Computing Machinery, pp 295–305, https://doi.org/10.1145/3351095.3372852

Zhang Z, Jaiswal P, Rai R (2018) FeatureNet: Machining feature recognition based on 3D convolution neural network. Computer-Aided Design 101:12–22. https://doi.org/10.1016/j.cad.2018.03.006